

What is a Package?

A package in Java is used to group related classes. Think of it as a **folder** in a file system. Packages prevent naming conflicts and make code searching/access control much easier.

1. Built-in Packages (Java API)

Java provides a massive library of pre-written classes. To use them, we use the `import` keyword.

```
import java.util.Scanner; // Import single class
import java.util.*;      // Import entire package
```

Example: Handling user input with the Scanner class.

```
Scanner myObj = new Scanner(System.in);
String name = myObj.nextLine();
```

2. User-Defined Packages

To create your own package, use the package keyword at the very top of your source file.

```
└─ root
  └─ mypack
    └─ MyClass.java
```

```
package mypack;

class MyClass {
    public static void main(String[] args) {
        System.out.println("Inside mypack!");
    }
}
```

Key Rules

- **Naming:** Package names should be lowercase to avoid conflicts with class names.
- **Importing:** Use `import package.name.ClassName` to use code from elsewhere.
- **Compilation:** Use `javac -d . FileName.java` to compile the package structure automatically.